Prototype-as-Prompt: Multimodal Sentiment Prototypes Endowing Large Language Models the Capability to Perform Multimodal Sentiment Analysis

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Abstract

Multimodal Sentiment Analysis (MSA) aims to integrate textual, acoustic, and visual information to predict sentiment polarity. With the emergence of Large Language Models (LLMs), existing studies commonly employ learnable queries to compress audio–visual representations and feed them as soft prompts into LLMs for MSA. However, due to the implicit learning mechanism of the learnable queries, these learnable queries lack explicit guidance regarding how each query encodes sentiment semantics. To address this issue, we propose a prototype-as-prompt framework that maps audio–visual representations into a fixed set of multimodal sentiment prototypes. These prototypes are then used as soft prompts to guide the LLM in performing MSA. Concretely, we first compress both textual and non-textual features into multimodal prototypes using a resampling-based strategy. We further introduce a sentiment-aware prototype learning that explicitly binds multimodal prototypes with sentiment semantics. To ensure both cross-modal consistency and intra-modal diversity of multimodal sentiment prototypes, we design a cross-modal prototype alignment constraint and a distance-weighted prototype diversity constraint. Extensive experiments across three LLMs and four benchmark datasets show that PaP achieves superior performance with only 0.09%–0.26% of trainable parameters, highlighting its effectiveness and parameter efficiency.

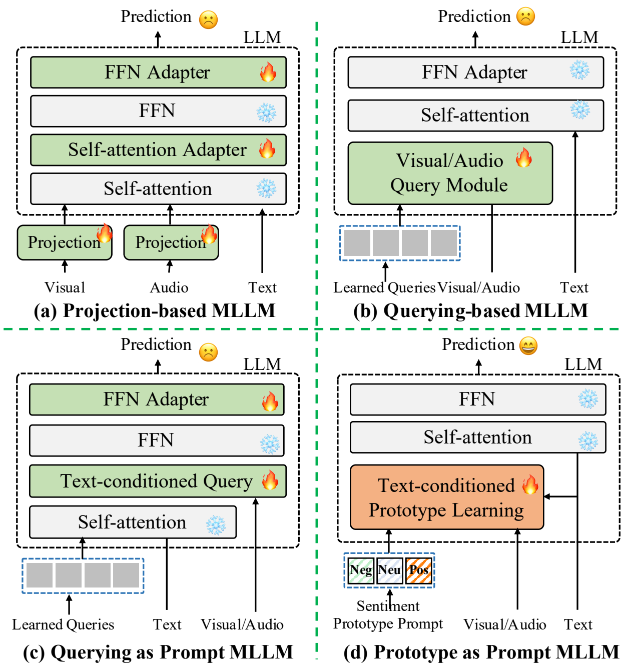

Figure 1. Four distinct approaches empower LLMs with multimodal sentiment analysis capabilities: (a) Projection-based method: Non-textual modalities are projected into the semantic space of the LLM, and an adapter is fine-tuned for alignment. (b) Query-based method: the resampler compresses non-textual modality information into learnable tokens. (c) Query-as-Prompt method: A text-conditioned resampler encodes non-textual modalities into learnable tokens that serve as prompts. (d) Prototype-as-Prompt (Ours): Non-textual modalities are mapped into a fixed number of sentiment prototypes, conditioned on the textual modality, which are then used as prompts.

1. Introduction

Multimodal Sentiment Analysis (MSA) is a fundamental yet challenging task [6, 30, 35, 41, 42, 46, 56] in affective computing, aiming to integrate multimodal signals, such as text, vision, and audio, for more accurate sentiment polarity recognition. Compared with unimodal inputs, multimodal signals contain richer sentiment cues, and effectively exploiting and fusing these cues is crucial for improving model performance. Traditional MSA and multimodal learning approaches mainly focus on constructing complex cross-modal interaction networks [28, 30, 35, 47, 48, 51, 52, 54–56] or designing alignment losses in geometric spaces [4–6, 10, 19, 25, 38, 43, 45, 46, 50, 53, 53] to enhance rep-

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resentational consistency across modalities.

Recently, the emergence of LLMs has brought a new paradigm to natural language processing and computer vision. With powerful reasoning and contextual understanding capabilities, LLMs have demonstrated impressive performance on a variety of downstream tasks. Fine-tuning LLMs on domain-specific datasets has become a popular approach, yielding strong results across multiple application areas. This success has motivated researchers to explore integrating non-textual modalities into LLMs for multimodal understanding, including sentiment analysis. As illustrated in Figure 1, existing efforts to incorporate non-textual modalities into LLMs for MSA can be broadly grouped into three categories: i) **Projection-based methods** (Fig. 1a): These approaches [3, 35, 44] project non-textual modality features into the embedding space of an LLM through auxiliary encoders, allowing unified multimodal representation learning. To preserve the LLM's general language ability, lightweight adapters are often introduced while keeping the backbone frozen. However, such methods require feeding full-length modality sequences into the model, leading to high computational cost and redundant information. ii) **Query-based methods** (Fig. 1b): These methods [1, 16, 32] employ independent resamplers (e.g., Q-Former) to compress modality features into a small number of learnable tokens that interact with the LLM. While they effectively filter out irrelevant signals, the parameter scale grows significantly as the number of modalities increases. iii) **Query as Prompt methods** (Fig. 1c): These approaches [24] treat learnable queries as prompts and employ text-conditioned resamplers to transform non-textual features into text-guided latent representations, reducing cross-modal interaction complexity. Although efficient, these methods fail to explicitly encode sentiment semantics within the prompts and often rely on additional adapters, increasing parameter overhead.

Despite learnable queries hold potential as prompts, they inherently lack explicit sentiment semantics due to their implicit learning mechanism, without clear guidance on how each learnable query encodes specific sentiment meanings. To address this issue, we propose Prototype-as-Prompt, which utilizes a fixed set of multimodal sentiment prototypes as soft prompts, explicitly encoding sentiment semantics into the corresponding emotion prototypes. This approach enables the frozen backbone LLM to perform multimodal sentiment analysis. Concretely, we first resample the text modality, then perform resampling on the visual and audio modalities to generate corresponding multimodal prototypes. Subsequently, we introduce sentiment-aware prototype learning, cross-modal prototype alignment, and distance-weighted prototype diversity constraints to bind explicit sentiment semantics to the multimodal prototypes. These sentiment prototypes, along with textual features, serve as input to the LLM for performing multimodal emotion analysis. The main contributions of this work are threefold:

- **Prototype as Prompt**: We map both textual and non-textual modalities into a fixed number of multimodal sentiment prototypes, which are injected into the LLM as prompts alongside textual inputs.
- **Sentiment-Aware Prototype Learning**: We explicitly encode sentiment semantics into a fixed set of multimodal prototypes, thereby creating discriminative multimodal sentiment prototypes as soft prompts for the LLM, guiding it to perform multimodal sentiment analysis.
- **Cross-Modal Prototype Alignment and Diversity**: We introduce a cross-modal distribution alignment constraint to ensure semantic consistency between visual and audio prototypes, together with a distance-weighted prototype diversity regularization that promotes expressive and diverse prototype learning within each modality.

2. Related Work

Multimodal sentiment analysis synergistically leverages multi-source signals from modalities such as text, vision, and audio for sentiment polarity prediction. Based on the network architecture design, existing multimodal sentiment analysis methods can be categorized into two types: 1) Multimodal Fusion, which designs complex fusion networks to integrate complementary multimodal data, and 2) Multimodal Representation Learning, which customizes diverse optimization losses to optimize single-modal or multimodal representations in the geometric space.

The former includes designs ranging from tensor-based fusion neural networks to attention mechanism-based neural networks [28, 29, 41, 44, 47], aiming to enhance the learning of multimodal information. Specifically, Li et al. [18] introduced Coupled Mamba, the first state-space model that couples multi-modal state chains to enable both intra-modal independence and inter-modal parallel interaction. Furthermore, Lv et al. [31] introduced a diffusion-driven dual-stream framework to improve multimodal sentiment analysis through enhanced feature fusion, and Jiang et al. [15] developed a decoupled dual-stream enhancement framework that improves modality fusion by a text-centric self-supervised mechanism. The latter focuses on unsupervised or semi-supervised contrastive learning to optimize single-modal and multimodal representations [6, 10, 20, 40, 42, 43, 45, 46, 50], such as learning decoupled multimodal representations and aligning cross-modal representations. Specifically, Jiang et al. [14] tackled modality heterogeneity and emotional inconsistency by optimizing representation learning and emotion discrimination jointly. Li et al. [20] further enhanced Mamba's cross-modal interaction modeling in multimodal fusion via local and global alignment, while Fang et al. [6] further addressed sample-level variability through dynamic weight fusion with distillation, balancing modality contributions and single-modality predictions.

With the development of LLMs, multimodal large language models (MLLMs) show strong potential for unified vision, audio, and language modeling. Existing methods can be mainly categorized into three types: (a) Projection-based MLLMs [3, 23, 33], which map multimodal features, such as visual or audio, into the text space through linear projection for modality fusion; (b) Querying-based MLLMs [1, 16, 32], which introduce query modules to learn cross-modal interactions and use learnable query vectors to retrieve relevant information from different modality feature spaces; and (c) Querying as Prompt MLLMs [24], where query results are used as prompts to guide the language model in reasoning. However, existing prompt designs are often heuristic or based on low-level features, making it difficult to fully capture semantic alignment across multi-modal features. Inspired by this paradigm, we compress the non-textual modal information into a fixed number of multimodal sentiment prototypes, explicitly encoding sentiment semantics into the corresponding multimodal sentiment prototypes. These prototypes are then used as soft prompts, input into the LLM to perform multimodal sentiment analysis.

3. Method

In this section, we present two customized modules, as illustrated in Fig. 2. Specifically, we first introduce the text-conditioned prototype construction in Section 3.1, followed by multimodal sentiment prototype learning in Section 3.2. Afterwards, we impose three constraints on the prototypes: sentiment-aware prototype learning, cross-modal prototype alignment, and distance-aware prototype diversity constraints. The objective function for optimizing the neural network is discussed in Section 3.3.

3.1. Text-Conditioned Prototype Construction

Bridging the cross-modal gap is one of the primary objectives of cross-modal alignment. Inspired by the Query-as-Prompt paradigm, we introduce sentiment prototypes as intermediaries to bridge the gap between different modalities. This strategy associates the textual representations from the LLM with features from non-text modalities through learnable prototypes. Given the textual modality $x_t \in \mathbb{R}^{T_t \times d}$, visual features $x_v \in \mathbb{R}^{T_v \times d}$, and audio features $x_a \in \mathbb{R}^{T_a \times d}$, along with the visual sentiment prototypes $P_v \in \mathbb{R}^{K \times d}$ and acoustic sentiment prototypes $P_a \in \mathbb{R}^{K \times d}$, we denote T_t, T_v and T_a as the sequence lengths of each modality, d as the feature dimension, and K as the number of sentiment prototype categories. We concatenate the textual features with the sentiment prototypes along the sequence dimension, and employ the self-attention mechanism to establish correlations between the textual tokens and the sentiment prototype tokens. Formally,

$$H = [P_v; P_a; x^t] \in \mathbb{R}^{(T_t+2K)\times d}, \tag{1}$$

$$[\tilde{P}_v, \tilde{P}_a, \tilde{x}_t] = \text{Softmax}\left(\frac{W_Q H H^\top {W_K}^\top}{\sqrt{d}}\right) W_V H, \tag{2}$$

where $\tilde{P}_v, \tilde{P}_a$ denotes the text-conditioned visual and acoustic prototypes. The symbol ';' denotes concatenation in sequence dimension. The W_Q, W_K, and W_V represent the learnable parameter.

3.2. Multimodal Sentiment Prototype Learning

Sentiment-aware Prototype Learning To integrate information from non-text modalities into their corresponding prototypes, we perform resampling on the non-text features using the text-conditioned visual and acoustic prototypes, generating new pseudo-tokens that effectively bridge the cross-modal gap. This process is implemented through a cross-modal attention mechanism. Specifically,

$$\hat{P}_v = \text{Softmax}\left(\frac{W_{Q_v} \tilde{P}_v {x_v}^\top {W_{K_v}}^\top}{\sqrt{d}}\right) W_{V_v} x_v, \tag{3}$$

$$\hat{P}_a = \text{Softmax}\left(\frac{W_{Q_v} \tilde{P}_a {x_a}^\top {W_{K_a}}^\top}{\sqrt{d}}\right) W_{V_a} x_a, \tag{4}$$

where $W_{Q_{\{a,v\}}}$, $W_{K_{\{a,v\}}}$, and $W_{V_{\{a,v\}}}$ represent the learnable parameter. Finally, the prototypes $\hat{P}_a$ and $\hat{P}_v$ along with the text representations, are incorporated into the LLM as prompts to perform sentiment polarity prediction.

Each prototype corresponds to a specific sentiment category, such as Strong Negative (SN), Negative (N), Weak Negative (WN), Neutral (Neu), Weak Positive (WP), Positive (P), and Strong Positive (SP) on the CMU-MOSEI dataset. To explicitly correlate each prototype with its corresponding sentiment semantics, the text-conditioned visual prototype $\hat{P}_v$ and acoustic prototype $\hat{P}_a$ are mapped through fully connected layers followed by activation functions to predict sentiment probability distributions:

$$h_v = \sigma(W_v \hat{P}_v + b_v), h_a = \sigma(W_a \hat{P}_a + b_a), \tag{5}$$

where W_v, W_a and b_v, b_a denote the learnable weight matrices and bias terms, and $\sigma(\cdot)$ represents the nonlinear activation function. The transformed features h_v and h_a are then fed into softmax classifiers to predict the corresponding sentiment categories:

$$\hat{y}_v = \text{Softmax}(h_v), \hat{y}_a = \text{Softmax}(h_a), \tag{6}$$

Finally, we adopt a cross-entropy loss to minimize the discrepancy between the predicted results $\hat{y}$ and the discrete ground-truth sentiment labels y:

$$\mathcal{L}_{semantic} = -\sum_{i=1}^{N} y_i (\log\hat{y}_{v_i} + \log\hat{y}_{a_i}). \tag{7}$$

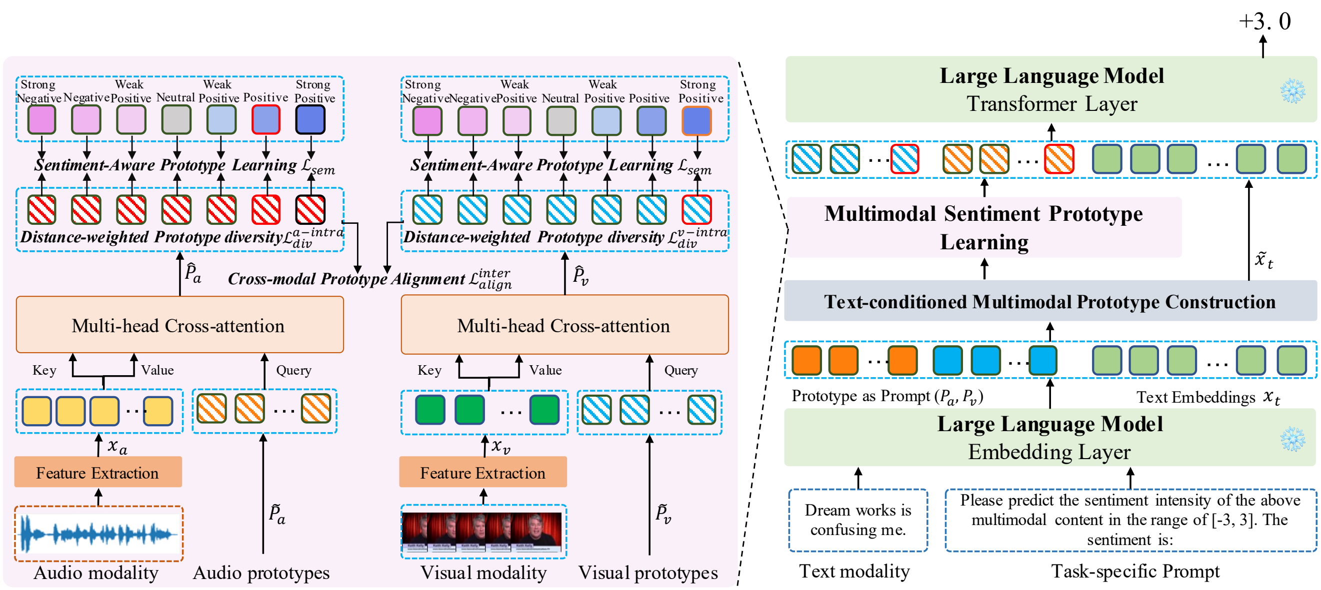

Figure 2. Schematic illustration of the proposed PaP framework. Text-conditioned Multimodal Prototype Construction resamples textual modality information into sentiment prototypes. Multimodal Sentiment Prototype Learning consists of three components: i) Sentiment-aware prototype learning, which explicitly binds sentiment semantics to each prototype; ii) Cross-modal prototype alignment, which aligns the distributions of prototypes across different modalities; and iii) Distance-weighted prototype diversity, which enhances intra-modal distinctiveness among sentiment prototypes. Finally, the learned prototypes serve as prompts, while textual features are used as inputs to the LLM for sentiment polarity prediction.

Cross-Modal Prototype Alignment To align the distribution of the text-conditioned visual and acoustic sentiment prototypes, we introduce a cross-modal prototype alignment loss based on the Central Moment Discrepancy (CMD) metric [49]. The loss encourages cross-modal sentiment prototypes to align semantic space by minimizing their order-wise moment differences. Let $\hat{P}_v$ and $\hat{P}_a$ denote the text-conditioned visual and acoustic prototype representations, with corresponding empirical distributions $\hat{p}_v$ and $\hat{p}_a$. The cross-modal prototype alignment loss is defined as:

$$\mathcal{L}_{align}^{inter} = CMD_{\mathcal{K}}(\hat{P}_v, \hat{P}_a), \tag{8}$$

where the $\mathcal{K}$-order central moment discrepancy on the interval $[a, b]^N$ is computed as:

$$\begin{aligned} CMD_{\mathcal{K}}(\mathcal{X}, \mathcal{Y}) &= \frac{1}{|b-a|}\|\mathbb{E}(\mathcal{X}) - \mathbb{E}(\mathcal{Y})\|_2 \\ &+ \sum_{\|=2}^{\mathcal{K}} \frac{1}{|b-a|^{\|}}\|C_k(\mathcal{X}) - C_k(\mathcal{Y})\|_2, \end{aligned} \tag{9}$$

$$\mathbb{E}(\mathcal{X}) = \frac{1}{|\mathcal{X}|}\sum_{x \in X} \mathcal{X}, \quad C_k(\mathcal{X}) = \mathbb{E}\big[(x - \mathbb{E}(\mathcal{X}))^k\big]. \tag{10}$$

Here, $\mathbb{E}(\mathcal{X})$ denotes the empirical mean vector of sample $\mathcal{X}$, and $C_k(\mathcal{X})$ is the k-th order central moment vector of its feature coordinates. The loss measures the discrepancy between the distributions of $\hat{P}_v$ and $\hat{P}_a$ by comparing their mean and higher-order central moments. As the loss decreases, the two prototype distributions become more consistent in the sentiment semantic space.

Distance-weighted Prototype Diversity. The text-conditioned visual and acoustic prototypes capture multimodal sentiment semantics from their respective modalities. However, when multiple prototypes within the same modality exhibit high similarity (e.g., weak positive, positive, and strong positive), redundant representations may arise, thereby diminishing the discriminative capacity of the sentiment prototypes for sentiment polarity differentiation. To address this issue, we introduce an orthogonality constraint loss on the sentiment prototypes, which encourages sufficient diversity among multimodal sentiment prototypes in the sentiment semantic space. This constraint effectively enhances the discriminative power of the learned sentiment prototypes by reducing redundancy and promoting orthogonal semantic representations. Let $\hat{P}_v \in \mathbb{R}^{K \times d}$ and $\hat{P}_a \in \mathbb{R}^{K \times d}$ denote the text-conditioned visual and acoustic prototypes, respectively. We define the orthogonal constraints for each modality as:

$$\mathcal{L}_{div}^{intra} = \sum_{m \in \{v, a\}} \lambda \left\| W \odot \left(\max(G_m, 0) - I_K\right)\right\|_F. \tag{11}$$

Here, $\mathbf{I}_K$ is identity matrices of sizes $\hat{P}_v$ and $\hat{P}_a$, $\mathbf{G}_m = \hat{P}_m(\hat{P}_m)^\top \in \mathbb{R}^{K\times K}, m \in \{v, a\}$, and $\|\cdot\|_F$ denotes the Frobenius norm. The $\lambda = \frac{1}{K(K-1)}$ and $W_{ij} = \frac{|i-j|}{K}$ denote the normalization term and the relative positional weight matrix, respectively. The $\max(\cdot, 0)$ operator ensures that prototype separation is enforced only when their cosine similarity is positive, thereby preventing excessive penalization of semantically close prototypes.

Next Token Prediction. The text features $\tilde{x}_t$, along with the two text-conditioned visual and prototypes $\{\hat{P}_v, \hat{P}_a\}$, are fed into the LLM to generate multimodal predictions:

$$\mathcal{L}_{task} = -\log p(y_i|\mathcal{I}, \theta), \tag{12}$$

where $\mathcal{I} = (\tilde{x}_t, \hat{P}_v, \hat{P}_a)$ and θ denotes the parameters of model.

3.3. Overall Objective

The final training objective combines all components into a joint optimization loss:

$$\mathcal{L} = \lambda_1\mathcal{L}_{task} + \lambda_2\mathcal{L}_{semantic} + \lambda_3\mathcal{L}_{align}^{inter} + \lambda_4\mathcal{L}_{div}^{intra}, \tag{13}$$

where λ_1, λ_2, λ_3 and λ_4 are hyper-parameters balancing the contributions of each term.

4. Experiment

4.1. Datasets

We use the following four datasets to evaluate the performance of the model: MOSEI [48], SIMS-V2 [27], MELD [34], and CHERMA [37].

MOSEI: MOSEI comprises 22,856 YouTube movie review clips annotated with sentiment scores ranging from -3 (extremely negative) to 3 (extremely positive). The videos span 250 topics and feature more than 1,000 unique speakers. Following standard splits, 16326 samples are used for training, 1,871 for validation, and 4,659 for testing.

SIMS-V2: SIMS-V2 consists of 4,403 curated video clips from various movies, TV shows, and variety programs, each labeled with an emotion score ranging from -1 (extremely negative) to 1 (extremely positive). The corpus is partitioned into 2,722 training, 647 validation, and 1,034 test instances.

MELD: MELD is a multimodal corpus curated for emotion recognition in conversational settings, comprising over 1,400 dialogues and 13,708 utterances from the TV series Friends. Each utterance is annotated with one of seven emotion categories: anger, disgust, fear, joy, neutral, sadness, and surprise.

CHERMA: CHERMA is a Chinese multimodal emotion recognition dataset comprising 28,717 clips extracted from 148 television dramas, 7 variety shows, and 2 films. And each utterance is annotated with one of seven emotion categories, consistent with MELD (anger, disgust, fear, joy, neutral, sadness, surprise).

Model	MOSEI				
	Acc-2	F1	Acc-7	MAE	Corr
TFN [47] [EMNLP17]	78.50	78.96	51.60	0.573	0.714
MulT [41] [ACL19]	81.15	81.56	52.84	0.559	0.733
MAG-BERT [35] [ACL20]	82.51	82.77	50.41	0.583	0.741
MISA [10] [ACMMM20]	83.60	83.80	52.20	0.555	0.756
Self-MM [45] [AAAI21]	82.81	82.53	53.46	0.530	0.765
MMIM [9] [EMNLP21]	82.24	82.66	54.24	0.526	0.772
CHFN [8] [ACMMM22]	83.70	83.90	54.30	0.525	0.778
UniMSE [12] [EMNLP22]	85.86	85.79	54.39	0.523	0.773
UniSA$_{\text{GPT2}}$ [22] [ACMMM23]	71.02	-	41.36	0.838	-
UniSA$_{\text{T5}}$ [22] [ACMMM23]	84.22	-	52.50	0.546	-
UniSA$_{\text{BART}}$ [22] [ACMMM23]	84.93	-	50.03	0.587	-
CMamba [18] [Nips24]	85.70	85.60	-	0.547	0.758
DDSE [15] [ACMMM25]	85.47	85.38	53.81	0.537	0.767
EMOE [6] [CVPR25]	85.50	85.50	53.90	0.530	-
AlignMamba [20] [CVPR25]	86.60	86.50	-	-	-
DiffuFuse [31] [ACMMM25]	86.70	86.40	55.40	0.521	0.776
MSE (Q) [44] [AAAI25]	84.12	83.45	52.02	0.558	0.725
MSE (L) [44] [AAAI25]	86.74	86.51	55.57	0.501	0.787
MSE (C) [44] [AAAI25]	86.91	86.77	54.56	0.515	0.783
PaP (Q) Ours	86.00	85.92	54.57	0.554	0.717
PaP (L) Ours	87.17	86.91	56.24	0.493	0.796
PaP (C) Ours	87.16	86.87	54.82	0.495	0.802

Table 1. Performance comparison on the MOSEI dataset. The symbols C, L, and Q denote ChatGLM-6B, LlaMA-2-7B, and Qwen-1.5B. Top three results are highlighted as best, second, and third, respectively.

4.2. Evaluation Metric

In our study, For MOSEI, each clip is annotated with a sentiment score ranging from -3 (Strong negative) to 3 (Strong positive); we use 7-class accuracy, binary accuracy (Acc-2), mean absolute error (MAE), Pearson correlation (Corr), and F1 score as evaluation metrics. For SIMS-V2, we evaluate binary accuracy (Acc-2), mean absolute error (MAE), Pearson correlation (Corr), F1 score, and Acc2_weak which measures performance on samples with sentiment intensity in the interval [-0.4, 0.4]. For MELD and CHERMA, since each utterance is labeled with one of seven emotions, we selected 7-class accuracy (Acc-7) and weighted F1 (WF1) as evaluation metrics.

Model	SIMS-V2				
	Acc-2	F1	Acc2.w	MAE	Corr
TFN [47] [EMNLP17]	76.51	76.31	66.27	0.323	0.667
MulT [41] [ACL19]	79.50	79.59	69.61	0.317	0.703
MAG-BERT [35] [ACL20]	79.79	79.78	71.87	0.334	0.691
MISA [10] [ACMMM20]	80.53	80.63	70.50	0.314	0.725
Self-MM [45] [AAAI21]	79.01	78.89	71.87	0.335	0.640
MMIM [9] [EMNLP21]	80.95	80.97	72.28	0.316	0.707
AV-MC [26] [ICMI22]	82.50	82.55	74.54	0.297	0.732
CMamba [18] [Nips24]	83.40	83.50	-	0.287	0.758
MSE (Q) [44] [AAAI25]	80.44	80.24	73.09	0.311	0.678
MSE (L) [44] [AAAI25]	75.53	75.44	68.61	0.382	0.553
MSE (C) [44] [AAAI25]	83.77	83.76	75.24	0.296	0.720
PaP (Q) Ours	81.23	81.25	74.23	0.303	0.687
PaP (L) Ours	76.23	76.25	70.50	0.376	0.550
PaP (C) Ours	84.75	84.80	76.18	0.264	0.769

Table 2. Performance comparison on the SIMS-V2 dataset. The symbols C, L, and Q denote ChatGLM-6B, LlaMA-2-7B, and Qwen-1.5B. Top three results are highlighted as best, second, and third, respectively.

Model	MELD		CHERMA	
	Acc	WF1	Acc	WF1
TFN [47] [EMNLP17]	60.77	57.74	-	68.37
MulT [41] [ACL19]	-	-	-	69.24
LFMIM [36] [ACL23]	-	-	-	70.54
MMGCN [13] [ACL21]	60.42	58.31	-	-
MM-DFN [11] [ICASSP22]	62.49	59.46	-	-
EmoCaps [21] [ACL22]	-	64.00	-	-
GA2MIF [17] [TAFFC23]	61.65	58.94	-	-
UniMSE [12] [EMNLP22]	65.09	65.51	-	-
UniSA$_{\mathrm{GPT2}}$ [22] [ACMMM23]	48.12	31.26	-	-
UniSA$_{\mathrm{T5}}$ [22] [ACMMM23]	64.52	62.17	-	-
UniSA$_{\mathrm{BART}}$ [22] [ACMMM23]	62.34	62.22	-	-
DRKF [14] [ACMMM25]	66.70	65.40	-	-
MSE (Q) [44] [AAAI25]	62.18	59.87	70.38	70.21
MSE (L) [44] [AAAI25]	65.14	63.66	71.58	71.41
MSE (C) [44] [AAAI25]	66.23	65.13	72.90	72.73
PaP (Q) Ours	63.30	61.38	73.93	73.87
PaP (L) Ours	66.54	65.46	72.70	72.65
PaP (C) Ours	67.47	65.97	75.33	75.28

Table 3. Performance comparison on the MELD and CHERMA datasets. The symbols C, L, and Q denote ChatGLM-6B, LlaMA-2-7B, and Qwen-1.5B. Top three results are highlighted as best, second, and third, respectively.

4.3. Experimental Settings

In the experiment, we adopt ChatGLM3-6B [7], Llama-2-7B [39], and Qwen-1.5B [2] as LLM bacbone, and the extraction of non-textual features is consistent with previous work [44]. All experiments are conducted with five random seeds [1111, 2222, 3333, 4444, 5555], and we report the mean over the five runs as the final result. We conduct experiments on V100 GPU with 32 GB memory. Training begins with a 30-epoch warm-up phase, and early stopping is triggered if the monitored validation metric (MAE) shows no improvement for 10 consecutive epochs. The hyperparameters λ_1, λ_2, λ_3, and λ_4 are set to 1, $1e^{-1}$, $1e^{-1}$, $5e^{-2}$, and $5e^{-2}$, respectively, based on the performance on the validation set. The batch size is set to 16.

4.4. Performance Comperision

To evaluate the effectiveness of our proposed PaP, we compare it with several state-of-the-art baselines on both multimodal sentiment analysis and multimodal emotion recognition tasks, such as TFN [47], MulT [41], MAG-BERT [35], MISA [10], Self-MM [45], MMIM [9], CHFN [8], UniMSE [12], DRKF [14], UniSA$_{\mathrm{GPT2}}$ [22], UniSA$_{\mathrm{T5}}$ [22], UniSA$_{\mathrm{BART}}$ [22], Coupled Mamba [18], DDSE [15], DiffuFuse [31], AlignMamba [20], EMOE [6], AV-MC [26], LFMIM [36], MMGCN [13], MM-DFN [11], EmoCaps [21], GA2MIF [17] and MSE [44]. The comparison results are summarized in Tables 1, 2, and 3. By analyzing the experimental results presented in these tables, we make the following observations: i) Benefiting from the strong representation capability of LLMs, the LLM-based method MSE achieves superior performance across multiple datasets, surpassing other baselines. This result demonstrates that freezing the LLM backbone while using textual features to filter non-textual features as prompts is an effective strategy. Our proposed PaP framework outperforms MSE across three LLMs and four benchmark datasets. This result indicates that mapping non-textual modalities into a fixed number of multimodal sentiment prototypes and feeding them as soft prompts into the LLM is an effective way to endow LLMs with multimodal sentiment analysis capability.

4.5. Ablation Study

To gain deeper insights into the effectiveness of the proposed components, we conduct incremental ablation studies. Specifically, we compare the PaP with the following variants: 1) w/o CPA, which removes the cross-modal prototype alignment constraint; 2) w/o SPL, which removes the

Variant Model	Acc-2	F1-Score
PaP (C)	**75.33**	**75.28**
w/o SPL	73.83	73.73
w/o CPA	74.01	73.74
w/o PDR	74.67	74.89
PaP (L)	**72.70**	**72.65**
w/o SPL	71.69	71.65
w/o CPA	72.01	72.30
w/o PDR	72.32	71.94
PaP (Q)	**73.93**	**73.87**
w/o SPL	72.74	72.27
w/o CPA	73.00	73.07
w/o PDR	73.47	73.38

Table 4. Ablation study results with different models on the CHERMA dataset.

sentiment-aware prototype learning loss; and 3) w/o PDR, which removes the distance-weighted prototype diversity regularization term.

The ablation results are illustrated in Figure 4. Compared to our proposed PaP, w/o CPA leads to a significant performance drop, for instance, the scores on metric Acc and F1-Score decrease by 1.32% and 1.54%, respectively. This finding validates the effectiveness of the cross-modal prototype alignment, which enables sentiment semantics to be aligned across modalities, ensuring that non-textual modalities convey consistent sentiment meaning, thereby enhancing the LLMs' sentiment analysis capability. Furthermore, the PaP outperforms the variant w/o SPL, revealing that introducing explicit sentiment supervision for prototype learning is crucial. Without such guidance, multimodal prototypes tend to lack clear sentiment semantics. Finally, the variant w/o PDR also shows degraded performance, indicating that incorporating diversity regularization into prototype learning is necessary, as it encourages different prototypes to capture distinct sentiment semantics, improving overall model robustness and representation capacity.

4.6. Sentiment Prototype Analysis

Beyond achieving superior performance, the key advantage of our proposed PaP framework lies in its ability to map varying modality information into the corresponding probability distributions of affective prototypes. To demonstrate this, we visualize the prototype probability distributions learned from both the visual and audio modalities, including samples annotated with positive and negative emotions. On the CMU-MOSEI dataset, we experiment with seven emotion prototypes: Strong Negative (SN), Negative (N), Weak Negative (WN), Neutral (Neu), Weak Positive (WP), Positive (P), and Strong Positive (SP). The results are shown in Figure 3.

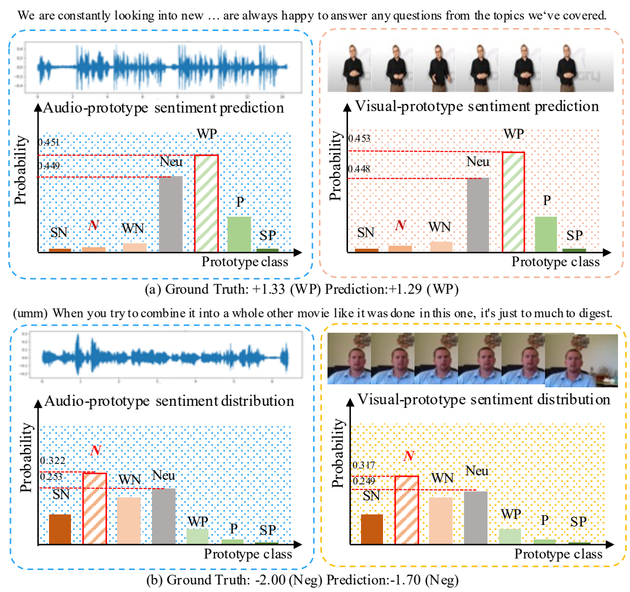

Figure 3. Visualization of sentiment prototype distributions for audio and visual modalities on CMU-MOSEI dataset.

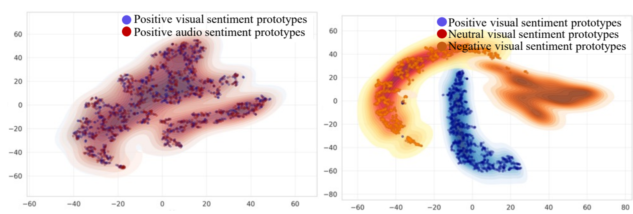

Figure 4. Visualization of the intra-modal diversity and cross-modal alignment of sentiment prototypes across audio and visual modalities.

As illustrated in Figure 3, our model effectively maps raw multimodal inputs into their corresponding sentiment prototypes. Specifically, in Figure 3, both the audio and visual modalities of samples labeled as Weak Positive are predominantly projected onto the Weak Positive prototype, which exhibits the highest probability among all categories. This observation validates the effectiveness of our proposed emotion-aware prototype learning, demonstrating its capability to associate non-textual modalities with the correct sentiment prototypes. Furthermore, the audio and visual modalities yield similar probability distributions, confirming the effectiveness of our cross-modal prototype alignment regularization. Finally, we observe that the probability distributions across different prototypes exhibit substantial divergence, suggesting that the prototype diversification regularization encourages better separation among sentiment prototypes, thereby facilitating the learning of pre-

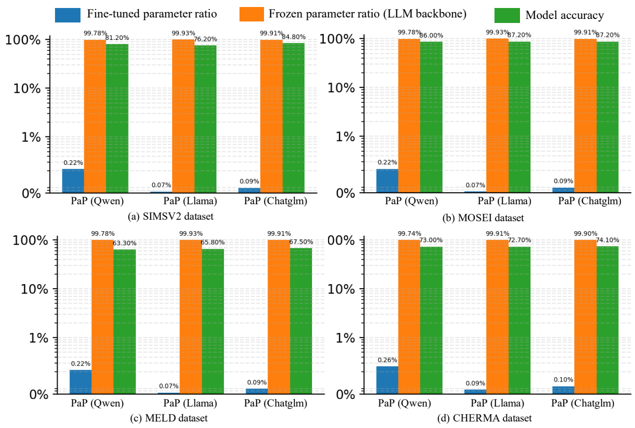

Figure 5. The proportions of trainable and frozen parameters, along with model performance, evaluated across three LLMs and four datasets.

cise affective semantics.

4.7. Prototype Alignment Visualization

To gain a deeper understanding of the proposed cross-modal prototype alignment module, we visualize the affective prototypes corresponding to the audio and visual modalities. Specifically, we project the prototypes of each modality for all samples in the dataset into a two-dimensional space using t-SNE. The resulting 2D embeddings are categorized into seven emotion prototypes: Strong Negative (SN), Negative (N), Weak Negative (WN), Neutral (Neu), Weak Positive (WP), Positive (P), and Strong Positive (SP). Each prototype group is visualized with a distinct color. The projection results are shown in Figure 4.

As illustrated in Figure 4 (left), the prototypes corresponding to the audio and visual modalities are highly entangled without clear boundaries, indicating that the cross-modal prototype alignment module effectively aligns the emotional semantics across modalities. Moreover, as shown in Figures 4 (right), there are distinct boundaries among the prototypes mapped to positive, negative, and neutral emotions, demonstrating that different affective prototypes capture semantically discriminative emotional information. These findings suggest that the non-textual affective prototypes learned from audio-visual modalities can serve as informative cues, thereby empowering LLMs with the capability to handle multimodal sentiment understanding.

4.8. Parameters Analysis

To investigate the impact of our proposed modules on the parameter efficiency of LLMs, we conduct experiments across three different LLMs and four multimodal sentiment analysis datasets. We report the proportion of parameters contributed by the PaP modules and the LLM backbones to the total number of model parameters. In our setup, only the parameters of PaP are trainable, while the LLM parameters remain frozen. The experimental results are illustrated in Figure 5.

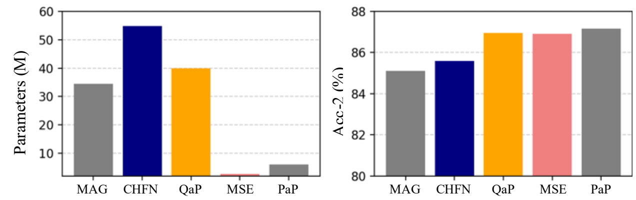

Figure 6. Trainable parameters and performance of different models (MAG, CHFN, QaP, MSE, PaP) on the CMU-MOSEI dataset.

As shown in Figure 5, the parameters of PaP account for only 0.09% to 0.22% of the total model parameters, which is extremely small compared to the frozen LLM parameters. Despite this small proportion, PaP achieves remarkable performance across all four multimodal sentiment analysis datasets. These results clearly demonstrate the high parameter efficiency of our proposed PaP framework and further validate the effectiveness of each module in our design.

To further assess the parameter efficiency and performance of the proposed model, we conduct a comparative study across four representative architectures: MAG, CHFN, QaP, and MSE. The experimental results are shown in Fig. 6. As illustrated in 6, MSE achieves the smallest parameter count, which is comparable to that of our PaP module, with both occupying a similarly low proportion of the overall model parameters. Notably, our model and MSE require substantially fewer parameters than MAG, CHFN, and QaP, while our PaP module simultaneously delivers the best performance among all baselines. These results collectively demonstrate the high parameter efficiency and superior performance of the proposed model.

5. Conclusion

In this paper, we propose a Prototype-as-Prompt (PaP) framework that endows LLMs with the capability to perform multimodal sentiment analysis. PaP first maps non-textual modalities into a fixed number of sentiment prototypes conditioned on the textual modality. Subsequently, PaP employs sentiment-aware prototype learning to bind sentiment semantics with their corresponding prototypes. By introducing cross-modal prototype alignment and a distance-weighted prototype diversity constraint, PaP enables the sentiment prototypes to capture high-level sentiment semantics. The sentiment prototypes, together with textual features, are then used as prompts to guide the LLM in predicting sentiment polarity. Extensive experiments conducted on four datasets and three LLMs demonstrate the effectiveness and flexibility of the proposed framework.

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